

JACK KILGORE

<https://jackilgore.com/>

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SUMMARY

I am a creative DSP programmer and composer with a reverence for all things related to sound. I have worked with artists and professional audio companies in designing and implementing music effects and synthesizers for the computer. I am interested in building technical and creative tooling for working with audio and musical structures. I prefer specificity and craftsmanship over generalization. I love music, I love tones, I love math!

PRIMARY EXPERIENCE

Lunacy Audio

2023 - 2026

Creative DSP Engineer and Programmer

- Creative real-time DSP engineer for designing and implementing professional-grade audio effects. Specialties include but are not limited to: reverb, distortion, resonators, filter banks, granulators, and pitch shifters. See: <https://lunacy.audio/products/beam/>.
- Lead designer of filter-bank resonator: [Bismuth](#) in collaboration with Virtual Riot.
- Lead designer of chorus effect: [Haze](#).
- Co-lead designer of multi-tap delay: [Taps](#) in collaboration with Benn Jordan.
- Co-lead designer of multi-distortion effect: [Volt](#).
- Co-lead designer of buffer shuffler effect: [Time](#).
- Real-time programming in C++.
- Continuous integration pipeline for building and codesigning Windows and Mac installers.
- General software engineering: architecture-design, optimization, git/github, etc.

Emission Control 2

2019 - 2023

Open Source Granular Synthesizer with Curtis Roads and Rodney DuPlessis

- Published article for MIT Press: Computer Music Journal 45:3.
[Architecture for Real-Time Granular Synthesis With Per-Grain Processing: EmissionControl2](#).
- Head software developer and designer
- Source code and manual can be found here: <https://github.com/EmissionControl2/EmissionControl2>
- Software can be found here: <https://github.com/EmissionControl2/EmissionControl2/releases>

Affection

2026

Self-Released Album

- 50 minute music composition for electronics and voice.
- [Full album and visualization](#)
- [Press](#)

A Special Mystic Joy

2025-Present

Music Software in Collaboration with Trevor Treglia

- Trevor Treglia composed a piece titled "A Special Mystic Joy" that employs frequency-dependent loopers. I was tasked with designing a piece of software for enabling the performance of the composition in real-time.

EDUCATION

University of California: Santa Barbara

2021 - 2023

Department of Media Arts and Technology
Masters of Science

University of California: Santa Barbara

2017 - 2021

Department of Computer Science
Bachelors of Science

UCSB Chancellor's Award for Excellence in Undergraduate Research (Engineering): [press](#)

TECHNICAL STRENGTHS AND TOOLS

- Cross-platform, real-time programming and signal processing
- Software engineering and architecture for maintaining professional-grade systems
- Git + Github power user
- Low-level knowledge on how to make computers run as fast as possible (profiler tooling, cache behavior, memory management, approximations, etc)
- Building cross-platform installers and continuous integration
- Deep listening and professional audio mixing
- Live performance and composition of computer music
- Programming languages: C++, Python, Javascript
- High level applications: Ableton, Reaper, Adobe products, Davinci Resolve
- Defensive and careful use of LLM tooling

SECONDARY EXPERIENCE

UCSB Computer Science TA

2021 - 2023

Teaching

- Formal Languages and Automata
- Data Structures and Algorithms
- Computer Graphics
- Human Computer Interaction

Assistant Engineer at KCSB

2018 - 2019

Occupation

- Soldering and fixing audio equipment.
- Building amplifier systems for live concerts.
- Live sound mixing and coordination of live events.

Eagle Scout

2016

Award

- Planned and coordinated a community service event for a local community center. Landscaped and cleaned up the lawn for the community events (AA meetings, daycare, etc.)
- Throughout my time as scout, I led multiple groups of people in wilderness training.

new

2026-Present

Music

- Music composition series studying the gap between computer and chamber music.

Dusk

2019

Music

- Music composition for electronics and field recordings performed at the International Computer Music Conference 2019 and the New York City Electroacoustic Music Festival.

20.8(3) Microseconds

2019-Present

Music

- Collaborative music project with Alex Meinhof
- [I'm Sorry About Last Night](#)
- The composition, "By the Broken Barn", was accepted to the International Computer Music Conference 2020 in Santiago, Chile.

Crest

2022-Present

Music

- A series of free music events in the Angeles Crest National Forest focusing on club music and real time deconstruction of club music.
- [Crest 1](#)
- [Crest 2](#)
- [Crest 3](#)

SBCAST Showcasing

2022-2023

- The Failed Songwriter (Documentation TBD)
- For Veronique ([Documentation](#))

Frame Feedback with P5.js

2021

- Frame feedback system implemented using p5.js and WebGL
- [code template for frame feedback](#)
- [example running in the browser](#)
- [high resolution render](#)
- [artistic spin](#)